

# Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub>: A High-Voltage Cathode for Sodium Batteries

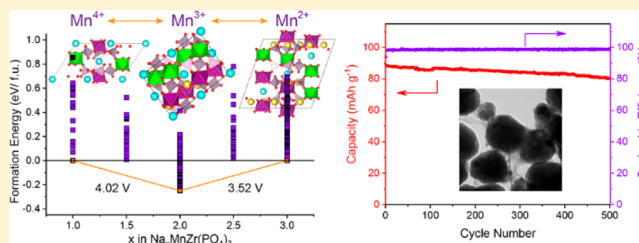
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## Supporting Information

**ABSTRACT:** Sodium batteries have been regarded as promising candidates for large-scale energy storage application, provided cathode hosts with high energy density and long cycle life can be found. Herein, we report NASICON-structured Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> as a cathode for sodium batteries that exhibits an electrochemical performance superior to those of other manganese phosphate cathodes reported in the literature. Both the Mn<sup>4+</sup>/Mn<sup>3+</sup> and Mn<sup>3+</sup>/Mn<sup>2+</sup> redox couples are reversibly accessed in Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub>, providing high discharge voltage plateaus at 4.0 and 3.5 V, respectively. A high discharge capacity of 105 mAh g<sup>-1</sup> was obtained from Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> with a small variation of lattice parameters and a small volume change on extraction of two Na<sup>+</sup> ions per formula unit. Moreover, Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> exhibits an excellent cycling stability, retaining 91% of the initial capacity after 500 charge/discharge cycles at 0.5 C rate. On the basis of structural analysis and density functional theory calculations, we have proposed a detailed desodiation pathway from Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> where Mn and Zr are disordered within the structure. We further show that the cooperative Jahn–Teller distortion of Mn<sup>3+</sup> is suppressed in the cathode and that Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> is a stable phase.



## INTRODUCTION

The integration of electric power generated from renewable sunlight and wind energy cannot be used unless it is stored in large-scale rechargeable batteries to smooth the supply of this electric power into the grid.<sup>1,2</sup> The ubiquitous availability of sodium from the oceans makes sodium batteries preferable to lithium batteries provided cathode hosts for Na<sup>+</sup> extraction/insertion can be identified with sufficiently high voltage versus Na<sup>+</sup>/Na, a long cycle life, and fast charge/discharge rates.<sup>3–5</sup> The stored energy density is  $\langle V(q) \rangle \times Q(I_{\text{dis}})$  at a constant current  $I_{\text{dis}} = dq/dt$ , where  $\langle V(q) \rangle$  is the cell voltage averaged over the state of charge  $q$  and the capacity  $Q(I_{\text{dis}})$  is the number of electrons transferred from the anode to the cathode per unit weight and/or volume in the time  $\Delta t$  for the electrochemical reaction at a desired rate between the two electrodes.<sup>6</sup> Because the electrodes change volume during a charge/discharge cycle, a small volume change of the cathodes during a charge/discharge cycle is preferred even with a liquid electrolyte.

A number of layered transition-metal oxides have been investigated as cathodes for a sodium battery, but they have been plagued by large volume changes and phase instabilities during cycling.<sup>7–9</sup> However, transition-metal cyanides with a Prussian-blue structure provide low-cost, facile synthesis and low activation energies for Na<sup>+</sup> insertion/extraction.<sup>10–14</sup> Another important category of Na<sup>+</sup> cathode hosts consists of polyanionic transition-metal compounds with open-framework

structures.<sup>15–19</sup> The strong binding in the polyanions can offer structural stability and a relatively small volume change during Na<sup>+</sup> insertion/extraction over a large solid-solution range.<sup>20,21</sup> Recently, several manganese-based polyanionic compounds have been shown to exhibit an interestingly high voltage versus sodium,<sup>22,23</sup> but an unacceptable performance has been attributed to Jahn–Teller distortions of Mn<sup>3+</sup> and/or a disproportionation reaction of surface Mn,  $2\text{Mn}^{3+} = \text{Mn}^{2+} + \text{Mn}^{4+}$ , with Mn<sup>2+</sup> dissolution into a liquid electrolyte.<sup>24</sup> Moreover, most of the polyanion hosts have used only the Mn<sup>3+</sup>/Mn<sup>2+</sup> couples; the limited number operating on the Mn<sup>4+</sup>/Mn<sup>3+</sup> couple have provided little interest. The Mn<sup>4+</sup>/Mn<sup>3+</sup> couple in Li<sub>2</sub>MnP<sub>2</sub>O<sub>7</sub> is located at a voltage that oxidizes conventional organic-liquid electrolytes.<sup>25</sup> The Mn<sup>4+</sup>/Mn<sup>3+</sup> couple is only partially (30%) active in Na<sub>3</sub>MnPO<sub>4</sub>CO<sub>3</sub> and cannot be accessed in Na<sub>4</sub>MnV(PO<sub>4</sub>)<sub>3</sub> from which the extraction of Na<sup>+</sup> ions proceeds in the sequences of the V<sup>4+</sup>/V<sup>3+</sup>, Mn<sup>3+</sup>/Mn<sup>2+</sup>, and V<sup>5+</sup>/V<sup>4+</sup> redox couples from the low to the high voltage.<sup>26,27</sup> Very little cycle stability was obtained in Na<sub>3</sub>MnTi(PO<sub>4</sub>)<sub>3</sub> and Na<sub>3</sub>Co<sub>0.5</sub>Mn<sub>0.5</sub>Ti(PO<sub>4</sub>)<sub>3</sub>.<sup>28,29</sup>

In order to reduce the concentration of Mn in a polyanion framework structure to where the cooperative Jahn–Teller distortion does not occur with all Mn<sup>3+</sup> and the surface disproportionation of the Mn<sup>3+</sup> to Mn<sup>2+</sup> and Mn<sup>4+</sup> might be

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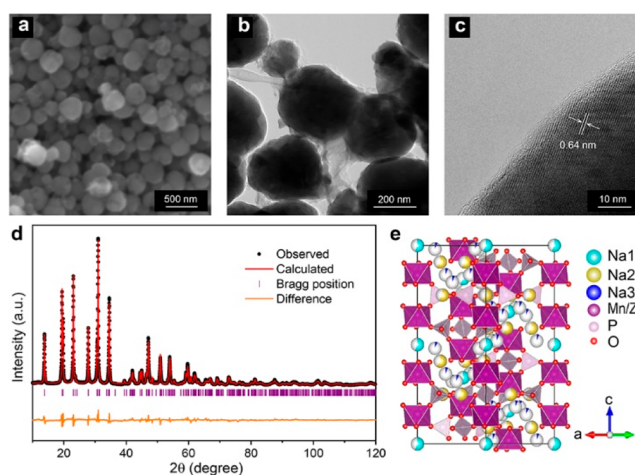
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suppressed, we have investigated as a sodium cathode  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  having the rhombohedral NASICON structure. In an early report, the measurement of ionic conductivity of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  revealed that the rhombohedral phase of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  has an activation energy of 0.38 eV, which is similar to the activation energies of rhombohedral phases of  $\text{Na}_3\text{V}_2(\text{PO}_4)_3$  (0.34 eV) and  $\text{Na}_3\text{Cr}_2(\text{PO}_4)_3$  (0.39 eV) and considerably smaller than the activation energies of the monoclinic phases of  $\text{Na}_3\text{V}_2(\text{PO}_4)_3$  (0.87 eV) and  $\text{Na}_3\text{Cr}_2(\text{PO}_4)_3$  (0.62 eV), suggesting its potential suitability as a cathode for sodium batteries.<sup>30–33</sup> However, in this early study,  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  was not investigated electrochemically as a cathode material of a sodium battery.<sup>30</sup> A subsequent study failed to obtain a reversible extraction of  $\text{Na}^+$  from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  synthesized by solid-state reaction and uncoated by carbon or any other electronic conductor, which has led to the suggestion that the problem is an instability of the octahedral-site  $\text{Mn}^{3+}$  ion.<sup>33</sup> Such an instability could be due to a cooperative Jahn–Teller distortion or to a surface disproportionation  $2\text{Mn}^{3+} = \text{Mn}^{2+} + \text{Mn}^{4+}$  with  $\text{Mn}^{2+}$  dissolution in the liquid electrolytes. The failure to extract  $\text{Na}^+$  reversibly was the result of a solid-state synthesis that gives micron-sized crystallites and a failure to coat the active electrode material that is an electronic insulator with a carbon coat, or a coat of another electronic conductor that also allows  $\text{Na}^+$  cross over.<sup>34–36</sup> In this paper, we have used a sol–gel synthesis to obtain 200 nm particles of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  that were coated *in situ* with a thin carbon layer. With these particles, we were able to extract/insert  $\text{Na}^+$  ions reversibly over a large solid-solution range, accessing not only the  $\text{Mn}^{3+}/\text{Mn}^{2+}$  redox couple but also the  $\text{Mn}^{4+}/\text{Mn}^{3+}$  redox couple at, respectively, 3.5 and 4.0 V versus  $\text{Na}^+/\text{Na}$ . Moreover, the cathode showed no cooperative Jahn–Teller distortion down to lowest temperatures and it can be cycled stably over many cycles indicating no evidence of surface disproportionation and  $\text{Mn}^{2+}$  dissolution. Moreover, we report a theoretical analysis of the  $\text{Na}^+$ -ion extraction sites in the rhombohedral NASICON-structured cathode  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  containing two quite different octahedral-site transition-metal cations whereas earlier theoretical studies have been confined to NASICON structures with only one octahedral-site transition-metal cation. The first-principles calculations also demonstrate the intrinsic stability of the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  structure during desodiation in addition to the changes in the local Mn environment.

## RESULTS AND DISCUSSION

**Synthesis and Crystal Structure of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ .** A sol–gel method was used to prepare the cathode material  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ ; the resulting material had a particle size of about 200 nm (Figure 1a). A thin layer of carbon was coated on the surface of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  particles during the synthesis procedure to provide electron tunneling to the surface  $\text{Na}^+$  ions with subsequent oxidation of the  $\text{Mn}^{2+}$  to  $\text{Mn}^{3+}$  and, eventually, to  $\text{Mn}^{4+}$  (Figure 1b); the amount of coated carbon was about 7.1% according to thermogravimetric analysis (Figure S1). The crystalline region of the cathode material exhibited lattice fringes with an interplanar spacing of about 0.64 nm, consistent with the (012) atomic planes of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  with the rhombohedral structure (Figure 1c).

A Rietveld refinement of the X-ray diffraction (XRD) pattern further confirmed the formation of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  with the rhombohedral NASICON structure in the space group  $R\bar{3}c$  (Figure 1d).<sup>30,37</sup> The absence of additional peaks



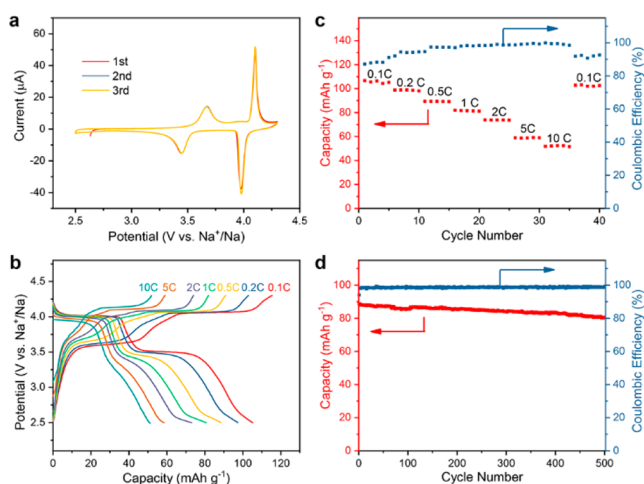
**Figure 1.** (a) SEM image, (b) TEM image, (c) high-resolution TEM image, (d) XRD pattern and Rietveld refinement, and (e) crystal structure of the cathode material  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ .

corresponding to long-range ordering between Mn and Zr in the XRD pattern indicates the octahedral sites were randomly occupied by the two kinds of transition metals in the NASICON framework.<sup>38,39</sup> The lattice parameters of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  ( $a = 8.988$  Å, and  $c = 22.594$  Å) are larger than those of the well-documented  $\text{Na}_3\text{V}_2(\text{PO}_4)_3$  cathode for sodium batteries ( $a = 8.738$  Å, and  $c = 21.815$  Å),<sup>40</sup> which provides sufficient interstitial volume for extraction/insertion of  $\text{Na}^+$  ions. The Rietveld refinement revealed the occupancy factors of 0.558, 0.596, and 0.066 for Na(1), Na(2), and Na(3) sites, respectively, corresponding to  $16.45$  ( $6 \times 0.558 + 18 \times 0.596 + 36 \times 0.066$ )  $\text{Na}^+$  ions per unit-cell ( $Z = 6$ ), consistent with the expected chemical formula  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  (Table S1).

The crystal structure of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  is composed of  $\text{MnO}_6$  and  $\text{ZrO}_6$  octahedral units bridged by corner-sharing  $\text{PO}_4$  tetrahedral units to establish a framework structure with interstitial sites accommodating three different types of  $\text{Na}^+$  ions: (i) the Na(1) sites with 6-fold coordination, (ii) the Na(2) sites with 8-fold coordination, and (iii) the Na(3) sites which are four-coordinate ( $\text{NaO}_4$ ) sites situated in-between the Na(1) and Na(2) sites (Figure 1e). An occupancy of Na(1) sites less than unity in the Rietveld refinement suggested that a portion of  $\text{Na}^+$  ions at Na(1) sites are displaced to the Na(3) sites, which has been observed in similar NASICON structures and suggests that simultaneous occupation of neighboring Na(1) and Na(2) sites by  $\text{Na}^+$  ions results in a Coulombic repulsion that displaces a  $\text{Na}^+$  ion from the Na(1) site to a Na(3) site.<sup>41</sup> The electrochemical activities of NASICON-structured materials are typically from extraction/insertion of  $\text{Na}^+$  ions located at Na(2) sites.<sup>42,43</sup> As is detailed in the following sections, the energies of the Na(1) and Na(2) sites are affected by the local Mn and Zr ordering. A theoretical capacity of  $107 \text{ mAh g}^{-1}$  could be obtained for  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  through utilization of both the  $\text{Mn}^{3+}/\text{Mn}^{2+}$  and  $\text{Mn}^{4+}/\text{Mn}^{3+}$  redox couples with extraction/insertion of two  $\text{Na}^+$  ions at Na(2) sites per formula unit.

**Electrochemical Performance of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ .** Two oxidation peaks at 3.6 and 4.1 V are revealed in the cyclic voltammetry profiles of the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  cathode, corresponding to the reversible extraction of two  $\text{Na}^+$  ions accompanied by the successive two-phase transformations

from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  to  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and then to  $\text{NaMnZr}(\text{PO}_4)_3$  (Figure 2a). The two reduction peaks located



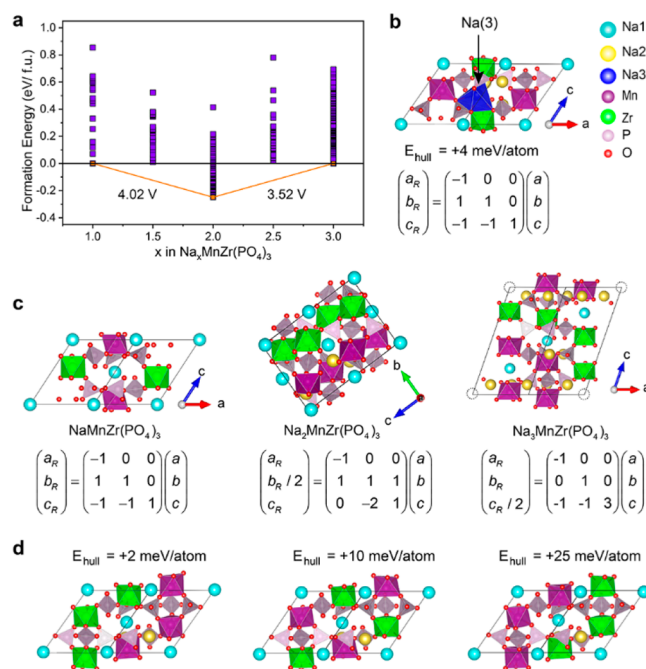
**Figure 2.** Electrochemical performance of the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  cathode. (a) Cyclic voltammetry profiles at a scan rate of  $1 \text{ mV s}^{-1}$ . (b) Charge/discharge profiles at rates from 0.1 to 10 C. (c) Discharge capacities and Coulombic efficiencies at rates from 0.1 to 10 C. (d) Charge/discharge cycling performance at 0.5 C rate.

at 4.0 and 3.5 V can be attributed to the successive  $\text{Mn}^{4+}/\text{Mn}^{3+}$  and  $\text{Mn}^{3+}/\text{Mn}^{2+}$  redox couples, accompanied by the transformations from  $\text{NaMnZr}(\text{PO}_4)_3$  to  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and then back to  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ . The charge profile of the cathode exhibits a voltage plateau at 3.6 V corresponding to the extraction of the first  $\text{Na}^+$  ion through the  $\text{Mn}^{3+}/\text{Mn}^{2+}$  redox couple followed by another voltage plateau at 4.1 V corresponding to the extraction of the second  $\text{Na}^+$  ion through the  $\text{Mn}^{4+}/\text{Mn}^{3+}$  redox couple (Figure 2b). Upon discharge, the two plateaus located at 4.0 and 3.5 V represent the insertion of two  $\text{Na}^+$  ions through the  $\text{Mn}^{4+}/\text{Mn}^{3+}$  and  $\text{Mn}^{3+}/\text{Mn}^{2+}$  redox couples, respectively. As a cathode material for sodium batteries,  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  delivers discharge voltage plateaus higher than the well-documented NASICON compounds  $\text{Na}_3\text{V}_2(\text{PO}_4)_3$  (3.4 V) and  $\text{NaTi}_2(\text{PO}_4)_3$  (2.1 V).<sup>44,45</sup> Moreover, the  $\text{Mn}^{4+}/\text{Mn}^{3+}$  redox couple in  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  provides a discharge voltage (4.0 V) higher than the well-documented  $\text{Mn}^{3+}/\text{Mn}^{2+}$  redox couple in other manganese phosphate cathodes, such as  $\text{Na}_2\text{MnP}_2\text{O}_7$  (3.6 V) and  $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$  (3.8 V) for sodium batteries.<sup>46–48</sup>

The discharge capacity of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  was  $105 \text{ mAh g}^{-1}$  at 0.1 C rate, which is close to the theoretical capacity ( $107 \text{ mAh g}^{-1}$ ) corresponding to extraction/insertion of two  $\text{Na}^+$  ions per formula unit (Figure 2c). When the rate increased to 1 and 10 C, the discharge capacity of the cathode was 82 and  $52 \text{ mAh g}^{-1}$ , respectively. The low Coulombic efficiency at the low rate ( $\sim 90\%$  at 0.1 C) is most probably from the oxidation of the liquid electrolyte at the high charging voltage (4.3 V);<sup>49–51</sup> however,  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  still exhibits an initial Coulombic efficiency (87%), higher than that of other manganese phosphate cathodes for sodium batteries, such as  $\text{Na}_2\text{MnP}_2\text{O}_7$  (69%),  $\text{Na}_4\text{Mn}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$  (60%), and  $\text{Na}_2\text{MnPO}_4\text{F}$  (80%).<sup>48,52,53</sup> When the current density went back to the initial 0.1 C rate, the discharge capacity of the cathode returned to the value of  $104 \text{ mAh g}^{-1}$ , demonstrating a useful rate capability of the cathode. In addition, an excellent cycling stability was obtained for  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ ; 91% of the

initial capacity could be retained after 500 cycles at 0.5 C rate with an average Coulombic efficiency of 98.7% (Figure 2d). In terms of discharge voltage, reversible capacity, and cycling stability,  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  gives a performance superior to that of other manganese phosphate cathodes for sodium batteries reported in the literature (Table S2). The XRD pattern and TEM image of the cathode after the charge/discharge cycling tests were almost identical to those of the pristine one (Figure S2), which implies that the crystal structure of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  was well-maintained after repeated extraction/insertion of  $\text{Na}^+$  ions.

**Desodiation Mechanism of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ .** In order to understand further changes in the structure during desodiation, the energies of  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  structures from  $x = 1–3$  were computed with DFT. The convex hull shows the formation energy per formula unit ( $E_{\text{form}}$ ) of  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  phases relative to the end member compositions of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  ( $E_{\text{Na}_3}$ ) and  $\text{NaMnZr}(\text{PO}_4)_3$  ( $E_{\text{Na}_1}$ ) according to the formula  $E_{\text{form}} = E_{\text{Na}_x} - 1/2(x-1)E_{\text{Na}_3} - 1/2(3-x)E_{\text{Na}_1}$  (Figure 3a).



**Figure 3.** (a) Convex hull of  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  from DFT calculations. (b) Near hull DFT relaxed structure of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  containing a  $\text{Na}^+$  ion in a Na(3) site. (c) Structures of the lowest energy phase of  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and the relationship between the unit cell parameters ( $a$ ,  $b$ ,  $c$ ) of the DFT cells and the corresponding unit cell parameters of the  $R3c$  structure ( $a_R$ ,  $b_R$ ,  $c_R$ ). Na(1) vacancies are shown as dashed circles. (d) DFT optimized structures of  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  with different Mn/Zr orderings.

$\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  is a stable phase on the hull between the end member configurations, and the intermediate configurations at  $\text{Na}_{1.5}\text{MnZr}(\text{PO}_4)_3$  and  $\text{Na}_{2.5}\text{MnZr}(\text{PO}_4)_3$  are metastable/unstable. Two-phase reactions between  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  and  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and between  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and  $\text{NaMnZr}(\text{PO}_4)_3$  are therefore preferred, in accordance with XRD analysis in the following section. From the lowest energy structures, sequential reactions from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3 \rightarrow \text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3 \rightarrow \text{NaMnZr}(\text{PO}_4)_3$  were predicted to occur at 3.52 and 4.02 V based on the approach of Aydinol et al.,<sup>54</sup> in good agreement with the



experimentally observed charge plateaus at 3.6 and 4.1 V. The energy of the  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  structures on the hull ( $x = 1, 2$ , and 3) were compared to all phases in the Na–Mn–Zr–P–O phase diagram in the Materials Project,<sup>55</sup> and all of the structures were found to be stable with respect to decomposition to neighboring phases of  $\text{NaMnPO}_4$ ,  $\text{NaZr}_2(\text{PO}_4)_3$ ,  $\text{Na}_4\text{P}_2\text{O}_7$ ,  $\text{NaPO}_3$ ,  $\text{MnPO}_4$ , and  $\text{ZrO}_2$ , highlighting the stability of the desodiated  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  structure.

For the end member composition of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ , the small energy difference between the structures above the hull predicated with DFT calculations ( $E_{\text{hull}} = 1\text{--}35$  meV/atom) suggests that there is significant disorder between the Mn and Zr sites within the structure. At the local scale, the distribution of Mn and Zr sites in  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  leads to three possible types of Na(1) site that share faces with either two  $\text{MnO}_6$ , two  $\text{ZrO}_6$  or one  $\text{MnO}_6$  and one  $\text{ZrO}_6$  octahedron along the  $c$ -axis. In the lowest energy structure of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ , the  $\text{MnO}_6\text{--Na(1)–MnO}_6$  and  $\text{MnO}_6\text{--Na(1)–ZrO}_6$  sites are occupied, whereas the  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  sites are vacant (Figure 3c). This result suggests that the unfavorable electrostatic interaction between  $\text{Zr}^{4+}$  and  $\text{Na}^+$  may locally destabilize  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  configurations in  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ . This finding is consistent with the structural refinement in which the occupancy of the Na(1) sites was less than unity. During structural optimization of several low energy structures,  $\text{Na}^+$  ions initially located in a  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  configurations relaxed to the intermediate four coordinated ( $\text{NaO}_4$ ) sites situated between the Na(1) and Na(2) sites (Figure 3b), in accordance with the Rietveld refinement. The formation energy of the structure was only 4 meV/atom above the lowest energy structure, which suggests that these Na(3) sites may act as additional low-energy positions to accommodate  $\text{Na}^+$  ions in the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  structure.

In the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  structures, the  $\text{Na}^+$  ions occupy the Na(2) sites that are face sharing with either two  $\text{Mn}^{2+}$  or one  $\text{Mn}^{2+}$  and one  $\text{Zr}^{4+}$  octahedron, as a result of the smaller electrostatic repulsion of  $\text{Mn}^{2+}\text{--Na}^+$  interactions compared with  $\text{Zr}^{4+}\text{--Na}^+$  interactions. The high energy,  $\text{ZrO}_6\text{--Na(2)–ZrO}_6$  local configurations are therefore preferential sites for Na(2) vacancies. For a random distribution of Mn/Zr ions, the  $\text{ZrO}_6\text{--Na(2)–ZrO}_6$  configurations account for 1/4 of the total Na(2) vacant sites. As approximately 1/3 of the total Na(2) sites are vacant in  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ , the remaining 1/12 of Na(2) vacancies will preferentially occur in  $\text{MnO}_6\text{--Na(2)–ZrO}_6$  configurations rather than in the  $\text{MnO}_6\text{--Na(2)–MnO}_6$  configurations, based on the previous electrostatic arguments and analysis of the low energy structures. The energies of the structures are also sensitive to the local Na ordering due to  $\text{Na}^+\text{--Na}^+$  interactions. However, the random distribution of Mn/Zr ions and high energy of the  $\text{ZrO}_6\text{--Na(2)–ZrO}_6$  sites precludes the formation of long-range Na orderings, such as that observed in the rhombohedral structure of  $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ .<sup>56</sup>

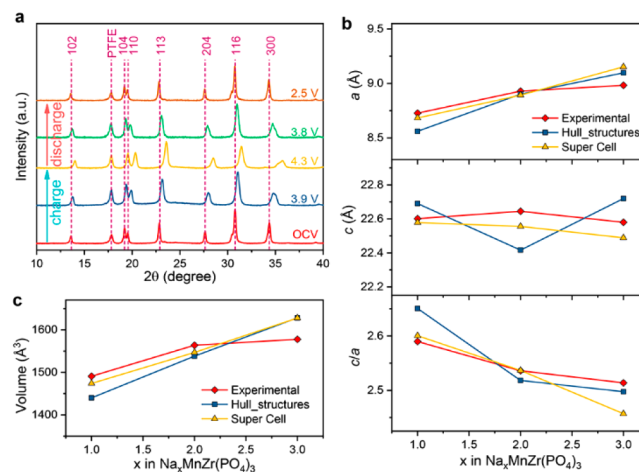
Analogous to  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ , there is a weak preference for  $\text{Mn}^{3+}/\text{Zr}^{4+}$  ordering in  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  (Figure 3d). While the lowest energy structure of  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  had fully occupied Na(1) sites, the Na(1) sites along  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  or  $\text{ZrO}_6\text{--Na(1)–MnO}_6$  configurations are also less stable than  $\text{MnO}_6\text{--Na(1)–MnO}_6$  configurations, and  $\text{Na}^+$  ions in  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  or  $\text{ZrO}_6\text{--Na(1)–MnO}_6$  configurations in several structures relaxed to Na(3) sites during the structural optimization, leading to vacant Na(1) sites along

either  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  or  $\text{ZrO}_6\text{--Na(1)–MnO}_6$  configurations.  $\text{Na}^+$  ions occupy 1/3 of the Na(2) sites in  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$ , and the site containing a  $\text{MnO}_6\text{--Na(2)–MnO}_6$  configuration will be preferentially occupied because of the lower energy than the site containing a Na(2) ion in a  $\text{MnO}_6\text{--Na(2)–ZrO}_6$  or  $\text{ZrO}_6\text{--Na(2)–ZrO}_6$  configuration.

For the  $\text{NaMnZr}(\text{PO}_4)_3$  end member, all of the Mn sites are oxidized to  $\text{Mn}^{4+}$ . In the lowest energy structure, the Na(1) sites are fully occupied whereas the Na(2) sites are unoccupied. In the small unit cell structures that initially contained vacancies in  $\text{MnO}_6\text{--Na(1)–MnO}_6$  or  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  configurations, adjacent  $\text{Na}^+$  ions in Na(2) sites relaxed to occupy the Na(1) sites. The lowest energy structure containing a  $\text{ZrO}_6\text{--Na(1)–MnO}_6$  vacancy was 18 meV/atom above the hull, suggesting that there is a strong preference for full occupancy of the Na(1) sites in  $\text{NaMnZr}(\text{PO}_4)_3$ .

From these results, a tentative mechanism for the desodiation of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  can be proposed. On initial desodiation from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  to  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  at 3.6 V,  $\text{Na}^+$  ions are preferentially extracted from the partially occupied  $\text{MnO}_6\text{--Na(2)–ZrO}_6$  sites while  $\text{MnO}_6\text{--Na(2)–MnO}_6$  sites remain fully occupied. The  $\text{Na}^+$  ions in local  $\text{MnO}_6\text{--Na(1)–MnO}_6$  environments remain fully occupied in both  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  and  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$ . The occupation of  $\text{ZrO}_6\text{--Na(1)–ZrO}_6$  sites increases on desodiation to  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$ , although displacement of Na(1) ions to Na(3) sites may also take place more easily due to the increase in the number of Na(2) vacancies. On further desodiation to  $\text{NaMnZr}(\text{PO}_4)_3$  at 4.1 V, the Na(1) sites become fully occupied, while the Na(2) sites are fully unoccupied.

**Structural Evolution of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ .** The phase transition of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  during extraction/insertion of  $\text{Na}^+$  ions was investigated by *ex situ* XRD analysis (Figure 4a).



**Figure 4.** (a) XRD patterns of the  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  cathode at different states of charge and discharge. (b) Variation of lattice parameters of  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  structures. (c) Variation of volume of  $\text{Na}_x\text{MnZr}(\text{PO}_4)_3$  structures.

A two-phase transformation from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  to  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  was observed on extraction of the first  $\text{Na}^+$  ion, followed by another two-phase transformation process from  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  to  $\text{NaMnZr}(\text{PO}_4)_3$  on extraction of the second  $\text{Na}^+$  ion. The phase transition of  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  is accompanied by the changes in the unit cell parameters

(Figure 4b), which are closely linked to the occupation of the Na(1) and Na(2) sites.<sup>57,58</sup>

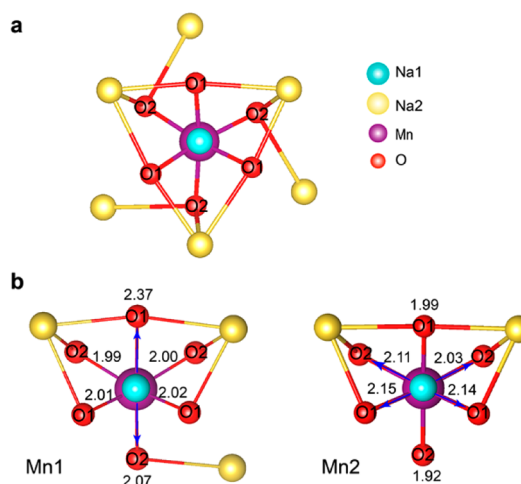
There is a small overall increase in the *c*-lattice parameter during desodiation (0.1%), in agreement with the DFT calculation (0.4%) from the supercell containing 8 formula units of Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> with disordered Mn and Zr sites (Figure S3). It is known from previous studies that the presence of Na(1) vacancies leads to an expansion of the *c*-axis.<sup>59</sup> Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> contains Na(1) vacancies at the ZrO<sub>6</sub>–Na(1)–ZrO<sub>6</sub> sites. On desodiation to Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub>, one of the Na<sup>+</sup> ions remained in a ZrO<sub>6</sub>–Na(1)–ZrO<sub>6</sub> site, while the other Na(1) site was empty as the Na<sup>+</sup> ion occupied an adjacent Na(3) site. On desodiation to NaMnZr(PO<sub>4</sub>)<sub>3</sub>, the Na(1) sites are filled and there is a small increase in the *c*-lattice parameter, which has previously been ascribed to an increased repulsion of the transition metals (i.e., Mn<sup>4+</sup>–Mn<sup>4+</sup> vs Mn<sup>2+</sup>–Mn<sup>2+</sup>) along the *c*-axis at higher states of charge. The small overall change in the *c*-lattice parameter observed experimentally during desodiation is related to the competing effects of transition metal repulsion causing expansion and increasing Na(1) site occupation causing contraction.

The small increase in the *c*-lattice parameter during desodiation is accompanied by a decrease in the *a*-lattice parameters and therefore an increase of *c/a* ratio. The decrease in the *a*-lattice parameter is commonly ascribed to a decrease in the Na(2) site occupation and a decrease in the effective ionic radius of the transition metal during oxidation, which for the current system are 0.83, 0.645, 0.53 Å for Mn<sup>2+</sup>, Mn<sup>3+</sup>, and Mn<sup>4+</sup>, respectively.<sup>60</sup> The overall decrease in the *a*-lattice parameter observed experimentally (2.8%) is smaller than predicted from the supercell (5.1%) and the lowest energy structures in the convex hull (5.9%), which may be due to the residual transition-metal ordering in the calculated structures in addition to the lack of temperature effects in the nominally 0 K DFT calculations. Moreover, Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> has a smaller volume change (5.5%) on desodiation to NaMnZr(PO<sub>4</sub>)<sub>3</sub> than other cathode materials, such as Na<sub>4</sub>Mn<sub>3</sub>(PO<sub>4</sub>)<sub>2</sub>P<sub>2</sub>O<sub>7</sub> (~7%) and Na<sub>3</sub>V<sub>2</sub>(PO<sub>4</sub>)<sub>3</sub> (~8.3%) for sodium batteries, and LiFePO<sub>4</sub> (~6.8%) for lithium batteries (Figure 4c).<sup>52,61,62</sup> In general, the small variation in the *a* and *c* lattice parameters and the small volume change of Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> is key to the long-term cycling stability.

The oxidation states of Mn and Zr in the Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> electrode was probed by X-ray photoelectron spectroscopy (XPS) at different states of charge (Figure S4). The binding energy of Mn 3p located at 48.8 eV before charging is in agreement with the presence of Mn<sup>2+</sup> in the pristine Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> electrode. The binding energy of Mn 3p increased to 49.5 eV after charging the electrode to 3.9 V, indicating the oxidation of Mn<sup>2+</sup> to Mn<sup>3+</sup>. At the end of charging the electrode to 4.3 V, the binding energy of Mn 3p further increased to 50.2 eV, implying the oxidation of Mn<sup>3+</sup> to Mn<sup>4+</sup>.<sup>63</sup> The binding energies of Zr 3d remained unchanged in the charging process, which suggests that the extraction/insertion of Na<sup>+</sup> ions proceeds through utilization of the Mn<sup>3+</sup>/Mn<sup>2+</sup> and Mn<sup>4+</sup>/Mn<sup>3+</sup> redox couples, keeping the valence state of Zr<sup>4+</sup> unchanged. The change in the oxidation state is further supported by calculation of the magnetic moment on the Mn and Zr sites of the lowest energy structures in the convex hull through a Bader analysis (Table S3).<sup>64</sup> The magnetic moment on Zr remains zero through-out the desodiation process and the charge remains approximately constant at +2.70 e, consistent with a fixed Zr<sup>4+</sup> oxidation state with a *d*-electron

configuration of *d*<sup>0</sup>. The change in the magnetic moment on Mn is consistent with the oxidation of high-spin Mn<sup>2+</sup> (*d*<sup>5</sup>) to Mn<sup>3+</sup> (*d*<sup>4</sup>) from Na<sub>3</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> → Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub>, followed by the oxidation of Mn<sup>3+</sup> to Mn<sup>4+</sup> (*d*<sup>3</sup>) from Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> → NaMnZr(PO<sub>4</sub>)<sub>3</sub>.

**Inhibited Cooperative Jahn–Teller Distortion of Manganese.** As all of the manganese ions in the Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> structure are in the Mn<sup>3+</sup> oxidation state, it may be expected that the octahedral MnO<sub>6</sub> centers would undergo a positive (negative) Jahn–Teller distortion in which two (four) of the Mn–O bond are lengthened and four (two) of the Mn–O bonds are shortened, as is observed in other Mn<sup>3+</sup> systems.<sup>65–67</sup> There are two distinct O sites in the rhombohedral NASICON structure: O1 and O2 (Figure 5a).



**Figure 5.** (a) Local bonding environment around MnO<sub>6</sub> site. (b) Coordination of two Mn sites, Mn1 and Mn2, in the lowest energy structure of Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub>. Mn–O bond lengths are shown in Å between the Mn sites and the two distinct O sites in the structure, O1 and O2. Jahn–Teller lengthened bonds are highlighted with blue arrows.

The O1 ions are located in between the Mn (or Zr) and Na(1) sites, forming the common faces between the MnO<sub>6</sub> and NaO<sub>6</sub> octahedra. The O2 sites are on the opposite side of the MnO<sub>6</sub> (or ZrO<sub>6</sub>) octahedra along the *c*-direction of the rhombohedral cell.

In the lowest energy Na<sub>2</sub>MnZr(PO<sub>4</sub>)<sub>3</sub> structure, MnO<sub>6</sub> centers have two different local environments (Mn1 and Mn2). For the Mn1 configuration, four of the Mn1–O bonds are shortened (S) from the average bond length of 2.08 Å and two Mn1–O bonds are lengthened (L), which is consistent with a positive Jahn–Teller distortion, however, there is significant deviation between the Mn1–O1 and Mn1–O2 bond lengths (Figure 5b). The Mn1–O1(L) bond, which shares a common O1 with two Na(2) sites, is considerably longer (2.37 Å) than the Mn1–O2(L) bond (2.07 Å) on the opposite site of the MnO<sub>6</sub> octahedron that shares a common O1 with one Na(2). The Mn1–O1(S) bonds in the plane perpendicular to the Mn1–O(L) bond are also slightly shorter than the Mn1–O2(S) bonds. The distribution of bond lengths is different for the Mn2 site; there are two Mn2–O(S) bonds and four Mn2–O(L) bonds, consistent with a negative Jahn–Teller distortion. As was the case for the Mn1 configuration, the Mn2–O1 bonds are longer than the Mn2–O2 bonds. Comparing the bond distributions of the two Mn sites, it

appears that the orientation of the local Jahn–Teller distortion is subtly related to the local Na ordering, in which there is a preference to maximize the number of Jahn–Teller lengthened bonds that share common oxygens with Na(2) sites. Although the  $\text{MnO}_6$  sites are distorted on the local scale, the lack of long-range ordering of the Na(2) sites suggests that a cooperative Jahn–Teller distortion of the  $\text{Mn}^{3+}$  centers as observed in other materials will be inhibited, which is beneficial for the long-term cycling stability.

## CONCLUSIONS

In conclusion, the NASICON-structured  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$  with three-dimensional channels for reversible extraction/insertion of  $\text{Na}^+$  ions is able to be cycled through a stable  $\text{Mn}^{3+}$  composition  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  to give access to both the  $\text{Mn}^{4+}/\text{Mn}^{3+}$  and  $\text{Mn}^{3+}/\text{Mn}^{2+}$  mixed-valent two-phase states. The octahedral sites at the  $\text{Mn}^{3+}$  are distorted by different local environments, not by a cooperative Jahn–Teller distortion that removes an orbital degeneracy. Moreover, a cooperative surface disproportionation reaction of  $\text{Mn}^{3+}$  is also suppressed, as there is no evidence of a  $\text{Mn}^{2+}$  dissolution into an organic liquid-carbonate electrolyte. On reversible extraction/insertion of two  $\text{Na}^+$  ions per formula unit, the voltage profile shows two plateaus, one for a two-phase  $\text{Mn}^{4+}/\text{Mn}^{3+}$  couple at 4.0 V and the other for a two-phase  $\text{Mn}^{3+}/\text{Mn}^{2+}$  couple at 3.5 V versus  $\text{Na}^+/\text{Na}$  with a step change at the  $\text{Mn}^{3+}$  composition  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$ . This high-voltage cathode for a sodium battery is superior to other manganese-phosphate cathodes reported in the literature, including an earlier study that failed to extract  $\text{Na}^+$  ions reversibly from  $\text{Na}_3\text{MnZr}(\text{PO}_4)_3$ . The stability of the  $\text{Mn}^{3+}$  phase  $\text{Na}_2\text{MnZr}(\text{PO}_4)_3$  and the local occupation of  $\text{Na}^+$  sites which was similar to that in NASICON structures with a single octahedral-site cation have been reproduced by DFT calculations. Nevertheless, still to be explored is the role of electron tunneling from/to a surface coat to/from surface  $\text{Na}^+$  ions in both the  $\text{Na}^+$  extraction/insertion reactions and the penetration of electrons between Mn ions separated by  $(\text{PO}_4)^{3-}$  anions.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.8b11388.

Experimental details; computational methodology; TGA of the cathode material; XRD pattern and TEM image of the cycled cathode; structure of supercells for DFT calculations; XPS spectra of the cathode at different charging state; crystallographic data of the cathode material; comparison with other cathode materials; magnetic moment and Bader charge calculations (PDF)

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## Notes

The authors declare no competing financial interest.

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